THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 13 & 14: **Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of 2, 4, 6, 8, 10 is: 1

(a) 8 **(b)** 5 **(c)** 6 **(d)** 30

2. The class mark of the class 30–40 is: 1

(a) 35 **(b)** 30 **(c)** 10 **(d)** 40

3. The empirical relation between mean, median and mode is: 1

(a) Mode = 3 Median − 2 Mean **(b)** Median = 3 Mode − 2 Mean

(c) Mean = 3 Median − 2 Mode **(d)** Mode = 2 Median − 3 Mean

4. A student states: "The modal class is the class with the highest class mark." The correct response is: 1

(a) Incorrect — it is the class with the highest frequency

(b) It is the last class

(c) Correct

(d) Cannot be determined

5. If every observation is increased by 5, Student A says the mean increases by 5; Student B says it is unchanged. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The probability of getting a head when a coin is tossed is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** 0 **(d)** 1

7. The probability of drawing a red card from a deck of 52 cards is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{13}$ **(d)** $\frac{13}{52}$

8. A die is thrown. The probability of a multiple of 3 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

9. A student states: "$P(E) + P(\text{not } E) = 0$." The correct response is: 1

(a) Incorrect — the sum is 1 **(b)** Correct

(c) The sum is 2 **(d)** Cannot be determined

10. A bag has 3 red and 7 white balls. Student A says $P(\text{white}) = \frac{7}{10}$; Student B says $\frac{7}{3}$. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of: 2

Class	0–10	10–20	20–30
Frequency	4	6	10

12. Find the mode of: 2

Class	10–20	20–30	30–40
Frequency	3	8	5

13. A box has 12 red and 8 blue balls. One is drawn at random. Find P(red) and P(blue). 2

14. Two coins are tossed. Find the probability of getting at most one head. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the median of: 3

Class	0–20	20–40	40–60	60–80	80–100
Frequency	6	10	15	12	7

16. A die is thrown twice. Find the probability that 3
(i) the sum is 7
(ii) the sum is more than 10
(iii) both numbers are the same.

17. Find the mean by the step-deviation method: 3

Class	10–30	30–50	50–70	70–90	90–110
Frequency	8	12	20	6	4

OR

Find the mode of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	2	5	9	7	3

18. Cards numbered 1 to 25 are in a box. One card is drawn. Find P(odd number), P(multiple of 3) and P(prime). 3

OR

A bag has 6 red, 4 blue and some green balls. If $P(\text{green}) = \frac{1}{3}$, find the number of green balls. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Marks of 60 students are given below. Find 5
(i) the mean
(ii) the median
(iii) the mode using the empirical relationship.

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Students	4	8	12	20	10	6

OR

The mean of the distribution is 25. Find p, and then find the median. 5

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	18	15	p	6

20. **Case study:** 500 raffle tickets are sold; there are 10 prizes of ₹100 and 2 prizes of ₹500. A person buys one ticket. Find 5
(i) P(any prize)
(ii) P(a ₹500 prize)
(iii) P(no prize).
(iv) A student says P(winning) $= \frac{1}{2}$ since one either wins or loses. Comment.

OR

Two dice are thrown. Find the probability that 5
(i) the sum is even
(ii) the product is even
(iii) the sum is a perfect square
(iv) a doublet of even numbers appears.

END OF QUESTION PAPER Total: 40 Marks